ModelIt!

Student Activity Pack

The Invisible Plant Factory

How Plants Build Themselves from Thin Air and Sunlight

Grade 6 | NGSS MS-LS1-6 | Energy and Matter: Flows, Cycles, and Conservation

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Where do you think plants get their food? Do they eat like we do?

2 If a tree weighs 10,000 pounds, where did all that weight come from?

3 Why are most plants green? Does the color matter?

4 Draw what you think happens inside a leaf when sunlight hits it.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sunlight Intensity | Water Availability | CO_2 Concentration | Chlorophyll Amount

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Perfect Day

High sunlight, adequate water and CO_2

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Cloudy Week

Lock Sunlight to 20%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Drought Stress

Water at 10%, other inputs normal

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Temperature — Enzymes that drive photosynthesis work best at specific temperatures — too hot or cold and they stop functioning

Soil Nutrients — Minerals like nitrogen, phosphorus, and potassium from soil are needed for building proteins and DNA, not for photosynthesis directly

Leaf Surface Area — More leaf surface area captures more sunlight, but also loses more water through stomata

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If plants make their own food from sunlight and air, where does the MASS of a giant tree actually come from? Use evidence from your simulation to support your answers.

1 Why is it surprising that most of a tree's mass comes from air instead of soil?

2 How are photosynthesis and cellular respiration opposite processes?

3 What would happen to Earth's oxygen supply if all plants died?

4 Why do plants grow toward light? What advantage does this give them?

5 How is a plant's leaf like a solar panel? How is it even better than a solar panel?

8 STEM Challenge: Design a Space Greenhouse

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YOUR ENGINEERING MISSION

Design a greenhouse for a Mars colony that provides maximum food production by optimizing photosynthesis conditions in an environment with limited resources.

THE SCENARIO

NASA is planning a Mars colony and needs a self-sustaining greenhouse. Mars has 95% CO_2 atmosphere but only 43% of Earth's sunlight, limited water, and no soil. Your agricultural engineering team must design a greenhouse using your photosynthesis model evidence.

Engineering Design Thinking

1. WHAT ARTIFICIAL LIGHTING SYSTEM WILL SUPPLEMENT THE WEAKER MARTIAN SUNLIGHT?
2. HOW WILL YOU SUPPLY AND RECYCLE WATER IN A CLOSED GREENHOUSE SYSTEM?
3. WHICH CROPS WILL YOU GROW TO MAXIMIZE CALORIES PER SQUARE METER?
4. HOW DOES YOUR GREENHOUSE DESIGN ADDRESS TEMPERATURE, PRESSURE, AND RADIATION ON MARS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Botanists and Agricultural Engineers develop new crop varieties, design sustainable farming systems, and study how to grow food in extreme environments — including space. They earn $55,000–$110,000/year.